W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). Enter 0.

8.15 Valve Component, VALVE

A valve junction component is indicated by VALVE for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the junction in the valve junction component is numbered CCC000000.

8.15.1 Valve Junction Geometry, CCC0101-0109

This card (or cards) is required for valve junction components.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

W3(R) Area (m^2, ft^2). This quantity is the full open area of the valve except in the case of a relief valve. For valves other than relief valves, if this area is input as 0, the area is set to the minimum area of adjoining volumes. If nonzero, this area is used. For relief valves, this term is the valve inlet throat area. If this term is input as 0, it will default to the area calculated from the inlet diameter term input on cards CCC0301-0309, in which case the inlet diameter term cannot be input as 0. If both this area and the inlet diameter are input as nonzero, this area will be used but must agree with the area calculated from the inlet diameter within 10^{-5} m^2. However, if this area is input as nonzero and the inlet diameter is input as 0, the inlet diameter will default to the diameter calculated from this area. When an abrupt area change model is specified, the area must be less than or equal to the minimum of the adjoining areas.

W4(R) Reynolds number independent forward flow energy loss coefficient, A_F. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section 8.15.3). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W6. If the smooth area change option is selected (a = 0), the entry here is the loss coefficient appropriate for the junction flow area resulting from the entry made in

W3. If one of the abrupt area change options is selected (a = 1 or 2), the entry here is the loss coefficient appropriate for the minimum flow area between the two hydrodynamic volumes connected by this junction. If the a = 1 option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section 8.15.3). See information for the previous word regarding interpretation and use of this loss coefficient.

W6(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the VALVE component, the junction control flag is limited to the format 0efvcahs

e-flag The digit e specifies the modified PV term in the energy equations. e = 0 means that the modified PV term will not be applied, and e = 1 means that it will be applied.

f-flag The digit f specifies CCFL options. f = 0 means that the CCFL model will not be applied, and f = 1 means that the CCFL model will be applied.

v-flag The digit v specifies horizontal stratification entrainment/pullthrough options. This model is for junctions connected to a horizontal volume. v = 0 means the model is not applied; v = 1 means an upward-oriented junction (offtake volume must be vertical); v = 2 means a downward-oriented junction (offtake volume must be vertical); and v = 3 means a centrally (side) located junction.

c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.

a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as ¨a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied. All options may be input for a motor or servo valve. If the smooth area change option is input, then a C_v table must be input; or, if no C_v table is input, then one of the abrupt area change options (a = 1 or 2) must be input. All options may be input for check valves, but if a nonzero leak ratio is entered, then one of the abrupt area change options (a = 1 or 2) must be input. All options may be input for a trip valve. For inertial and relief valves, one of the abrupt area change options (a = 1 or 2) must be input.

h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option; h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show h = 1.

s-flag The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to volume and the from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 uses momentum flux in the to volume, but not in the from volume. s = 3 does not use momentum flux in either the to or the from volume.

W7(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).

If the Henry-Fauske critical flow model is active, the discharge coefficient is entered. If W7 and W8 are missing, W7 is set to 1.0 and W8 is set to 0.14.

If the original RELAP5 critical flow model is active, the subcooled discharge coefficient is entered. This entry applies only to subcooled liquid choked flow calculations. The entry must be > 0.0 and < 2.0. If W7, W8 and W9 are missing, they are each set to 1.0.

W8(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).

If the Henry-Fauske critical flow model is active, the thermal nonequilibrium constant is entered. If W7 is entered and W8 is missing, W8 is set to 0.14. If the entry is < 0.01, the equilibrium option is used and the entry is reset to 0.0. If the entry is >1000.0, the frozen option is used and the entry is reset to 100.0.

If the original RELAP5 critical flow model is active, the two-phase discharge coefficient is entered. This entry applies only to two-phase choked flow calculations. The entry must be > 0.0 and < 2.0. If W7 is entered and W8 and W9 are missing, W8 and W9 are each set to 1.0.

W9(R) If the original RELAP5 critical flow model is active, the superheated discharge coefficient is entered. This entry applies only to superheated vapor choked flow calculations. The entry must be > 0.0 and < 2.0. If W7 and W8 are entered and W9 is missing, then W9 is set to 1.0. If the Henry-Fauske critical flow model is active, this word is not used and should not be entered.

8.15.2 Valve Junction Diameter and CCFL Data, CCC0110

This card is optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to specify the junction hydraulic diameter for the interphase drag calculation (i.e., f = 0 in Word 6 of cards CCC0101-0109), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If this card is being used for the CCFL model (i.e., f = 1 in Word 6 of cards CCC0101-0109), then enter all four words for the appropriate CCFL model if values different from the default values are desired.

W1(R) Junction hydraulic diameter, D_j (m, ft). This is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0. This number should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}}\right)$. If a 0 is entered or if the default is used, the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. See Word 3 of cards CCC0101-0109 for the area.

W2(R) Flooding correlation form, β. If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1. The default value is 0 (Wallis form).

W3(R) Gas intercept, c. This is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0. The default value is 1.

W4(R) Slope, m. This is the slope used in the CCFL correlation and must be > 0. The default value is 1.

8.15.3 Valve Junction Form Loss Data, CCC0111

This card is optional. The user-specified form loss is given in Words 4 and 5 of cards CCC0101-0109 if card CCC0111 is not entered. If card CCC0111 is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F Re^{-C_F}$$

$$K_R = A_R + B_R Re^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of cards CCC0101-0109. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R; the same information applies for B_F and B_R.

W1(R) B_F ($\geq$ 0). This quantity must be greater than or equal to 0.

W2(R) C_F ($\geq$ 0). This quantity must be greater than or equal to 0.

W3(R) B_R ($\geq$ 0). This quantity must be greater than or equal to 0.

W4(R) C_R ($\geq$ 0). This quantity must be greater than or equal to 0.

8.15.4 Valve Junction Initial Conditions, CCC0201

This card is required for valve junction components.

W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flows.

W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on the control word.

W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on the control word.

W4(R) Interface velocity (m/s, ft/s). Enter 0.

8.15.5 Valve Type, CCC0300

This card is required to specify the valve type.

W1(A) Valve type. This word must contain one of the variables from the following table.

Table 8.15-1 Valve Types

Valve Name	Type of Valve
CHKVLV	Check valve
TRPVLV	Trip valve
INRVLV	Inertial swing check valve
MTRVLV	Motor valve
SRVVLV	Servo valve,
RLFVLV	Relief valve

8.15.6 Check Valve Data and Initial Conditions, CCC0301-0399

These cards are required for a check valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The check valve behaves as an on, off switch. If the valve is on, then it is fully open; and if the valve is off, it is fully closed.

W1(I) Check valve type. Enter +1 for a static pressure-controlled check valve (no hysteresis), 0 for a static pressure/flow-controlled check valve (has hysteresis effect), or -1 for a static/dynamic pressure-controlled check valve (has hysteresis effect). It is recommended that 0 be used for most calculations, as it is more stable (i.e., less noisy and less oscillations) than +1 or -1.

W2(I) Check valve initial position. The valve is initially open if 0, closed if 1.

W3(R) Closing back pressure (Pa, lb_f/in^2). Additive closing ΔP to model spring-loaded valves.

W4(R) Leak ratio. This defines the flow area for leakage when the valve is nominally closed. If omitted or input as 0, then any of the area change models (a = 0, 1 or 2) may be specified for the check valve. If input as nonzero, then one of the abrupt area change models (a = 1 or 2) must be specified for the check valve. The leak ratio is specified as the fraction of the "junction area" through which leakage occurs. Because nonzero leak ratios require use of an abrupt area change model, the "junction area" is taken to mean the minimum of the flow areas in the two hydrodynamic volumes connected by the valve.

8.15.7 Trip Valve Data and Initial Conditions, CCC0301-0399

These cards are required for a trip valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The trip valve behaves as an on, off switch. If the valve is on, then it is fully open; and if the valve is off, it is fully closed.

W1(I) Trip number. This must be a valid trip number. If the trip is false, the valve is closed; if the trip is true, the valve is open.

8.15.8 Inertial Valve Data and Initial Conditions, CCC0301-0399

These cards are required for an inertial valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The inertial value behaves realistically in that the valve area varies considering the hydrodynamic forces and the flapper inertia, momentum, and angular acceleration. The abrupt area change model must be specified.

W1(I) Latch option. The valve can open and close repeatedly if the latch option is 0. When W1 = 1, the valve either opens or closes only once if the initial angle is between the maximum and minimum. If the flapper starts at either the maximum or minimum angle it will not move. When W1 = 2, the flapper will latch only at the maximum position. If it starts at the maximum, it will not move.

W2(I) Valve initial condition. The valve is initially open if 0, initially closed if 1.

W3(R) Cracking pressure (Pa, lb_f/in^2).

W4(R) Leakage fraction. Fraction of the "junction area" for leakage when the valve is nominally closed. Because an abrupt area change option must be used, the "junction area" is the minimum of the flow areas in the two hydrodynamic volumes connected by the valve.

W5(R) Initial flapper angle (degrees). The flapper angle must be within the minimum and maximum angles specified in Words 6 and 7.

W6(R) Minimum flapper angle (degrees). This must be greater than or equal to 0.

W7(R) Maximum flapper angle (degrees).

W8(R) Moment of inertia of valve flapper ($kg \bullet m^2$, $lb \bullet ft^2$).

W9(R) Initial angular velocity (rad/s).

W10(R) Moment length of flapper (m, ft).

W11(R) Radius of flapper (m, ft).

W12(R) Mass of flapper (kg, lb).

8.15.9 Relief Valve Data and Initial Conditions, CCC0301-0399

These cards are required for a relief valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The relief valve area varies, considering the hydrodynamic forces and the valve mass, momentum, and acceleration. One of the abrupt area change models (a =1 or 2) must be specified. The junction area input by cards CCC0101-0199 is the valve inlet area.

W1(I) Valve initial condition. The valve is initially closed if 0, open if 1.

W2(R) Inlet diameter (m, ft). This is the inside diameter of the valve inlet. If this term is input as 0, it will default to the diameter calculated from the junction area input on cards CCC0101-0109. If both this diameter and the junction area are input as nonzero, care must be taken that these terms are input with enough significant digits so that the areas agree within 10^{-5} m^2. If the junction area is input as 0, then this diameter must be input as nonzero.

W3(R) Valve seat diameter (m, ft). Nonzero input is required. This term is the outside diameter of the valve seat, including the minimum diameter of the inner adjustment ring. This term must also be greater than or equal to the inlet diameter.

W4(R) Valve piston diameter (m, ft). If input as 0, the default is to the valve seat diameter.

W5(R) Valve lift (m, ft). Nonzero input is required. This is the distance the valve piston rises above the valve seat at the fully open position.

W6(R) Maximum outside diameter of the inner adjustment ring (m, ft). If this input is 0, it will default to the valve seat diameter; in which case W7, following, must be input as 0. If this input is nonzero, the value must be greater than or equal to the valve seat diameter. If input is greater than the valve seat diameter, a nonzero input of W7, is allowed. Also refer to the warning stated for W9.

W7(R) Height of outside shoulder relative to the valve seat for inner adjustment ring (m, ft). Input of a positive, nonzero value is not allowed. Input of a 0 value is required if W6 preceding is defaulted or input equal to the valve seat diameter. If the shoulder is below the seat, this distance is negative. Also refer to the warning stated for W9.

W8(R) Minimum inside diameter of the outer adjustment ring (m, ft). If this input is 0, it will default to the valve piston diameter, in which case W9 must be input as positive and nonzero. If this input is nonzero, the value must be greater than or equal to the valve piston diameter. Input of a negative W9 is allowed only if this diameter is greater than the valve piston diameter. Also refer to the warning stated for W9.

W9(R) Height of inside bottom edge relative to the valve seat for outer adjustment ring (m, ft). This may be input as positive, 0, or negative. If this input is negative, then W8 preceding must be greater than the valve piston diameter. If the bottom edge is below the valve seat, this distance is negative. WARNING: Input of this term and terms W6, W7, and W8 preceding must be done with care to ensure that the resultant gap between the adjustment rings is positive and nonzero; otherwise, an input error will result.

W10(R) Bellows average diameter (m, ft). If this term is input as 0, it will default to the valve piston diameter, resulting in a model not containing a bellows for which the valve bonnet region is vented to the atmosphere.

W11(R) Valve spring constant (N/m, lb_f/ft). Positive, nonzero input is required.

W12(R) Valve setpoint pressure (Pa, lb_f/in^2). Positive input is required.

W13(R) Valve piston, rod, spring, bellows mass (kg, lb). Nonzero input is required.

W14(R) Valve damping coefficient (N•s/m, lb_f•s/ft).

W15(R) Bellows inside pressure (Pa, lb_f/in^2). Defaults to standard atmospheric pressure if omitted or input as 0.

W16(R) Initial stem position. This is the fraction of total lift and is required if W1 is input as 1. Total lift is input as W5.

W17(R) Initial valve piston velocity (m/s, ft/s). This must be 0 or omitted if W1 is input as 0

8.15.10 Motor Valve Data and Initial Conditions, CCC0301-0399

These cards are required for a motor valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The motor valve behaves realistically in that the valve area varies as a function of time by either of two models specified by the user. The user must also select the model for valve hydrodynamic losses by specifying either the smooth or the abrupt area change model. If the smooth area change model is selected (a=0), a table of flow coefficients must also be input as described in cards CCC0400-0499, CSUBV Table Section 8.15.12. If an abrupt area change model is selected (a = 1 or 2), a flow coefficient table cannot be input.

W1(I) Open trip number.

W2(I) Close trip number. Both the open and close trip numbers must be valid trips. When both trips are false, the valve remains at its current position. When one of the trips is true, the valve opens or closes depending on which trip is true. The transient will be terminated if both trips are true at the same time.

W3(R) Valve opening (or closing) change rate (s^{-1}). If Word 5 is not entered, this quantity is the rate of change of the normalized valve area as the valve opens (or closes). If Word 5 is entered, this quantity is the rate of change of the normalized valve stem position. If Word 6 is entered, this quantity is the rate of change of the normalized valve area as the valve opens only. This word must be greater than 0.

W4(R) Initial position. This number is the initial normalized valve area or the initial normalized stem position depending on Word W5. This quantity must be between 0.0 and 1.0.

W5(I) Valve table number. If this word is omitted or input as 0, the valve area is determined by the valve change rate and the trips. If this word is input as nonzero, the valve stem position is determined by the valve change rate and the trips; and the valve area is determined from a general table containing normalized valve area versus normalized stem position.

W6(R) Valve closing change rate (s^{-1}). This word is optional and if it is not entered, the valve closing change rate is the same as the valve opening change rate.

Input for general tables is discussed in cards 202TTTNN, General Table Data, Section 12. For this case, the normalized stem position is input as the *argument value* and the normalized valve area is input as the *function value*.

8.15.11 Servo Valve Data and Initial Conditions, CCC0301-0399

These cards are required for a servo valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The servo valve behaves as described for a motor valve except that the valve flow area or stem position is calculated by a control system. Input for control systems is discussed in Section 14. Input specifying the hydrodynamic losses for servo valves is also identical to that for motor valves.

W1(I) Control variable number. The value of the indicated control variable is either the normalized valve area or the normalized stem position, depending on whether Word 2 is entered. The control variable is also the search argument for the CSUBV table if it is entered.

W2(I) Valve table number. If this word is not entered, the control variable value is the normalized flow area. If it is entered, the control variable value is the normalized stem position, and the general table indicated by this word contains a table of normalized area versus normalized stem position. Input for the general table is identical to that for a motor valve.

8.15.12 Motor or Servo Valve CSUBV Table Factors, CCC0400

The CSUBV table may be input only for motor and servo valves. If the CSUBV table is input, the smooth area change model must be specified on the valve junction geometry cards (cards CCC0101-0109). If the smooth area change model is specified, a CSUBV table must be input.

This card is optional for a motor or servo valve. The factors apply to the flow area or the stem position and the flow coefficient entries in the CSUBV table.

W1(R) Normalized flow area or normalized stem position (See Section 8.15.10 for motor valve and Section 8.15.11 for servo valve).

W2(R) Flow coefficient factor.

8.15.13 Motor or Servo Valve CSUBV Table Entries, CCC0401-0499

These cards are required if the motor or servo valve requests a CSUBV table as part of its input. The CSUBV table contains forward and reverse flow coefficients as a function of normalized flow area or normalized stem position. These cards are required for a motor or servo valve.

The table is entered by using three-word sets. W1 is the flow area or stem position and must be normalized. The factor W1 on card CCC0400 can be used to normalize the flow area or stem position. In either case, the implication is that if the valve is fully closed, the normalized term is 0. If the valve is fully open, the normalized term is 1.0. Any value may be input that is between 0.0 and 1.0. The forward and reverse flow coefficients are W2 and W3, respectively. The code internally converts flow coefficients to energy loss coefficients by the formula $K = 2 \bullet A_j^2/(\rho \bullet CSUBV^2)$, where ρ is a reference density of 991.091 kg/m^3 (62.3706 lb$_m$/ft^3), A_j is the full open valve area, and CSUBV is the flow coefficient. On card CCC0400, W2 may be used to modify the definition of CSUBV. A smooth area change must be specified in W6 on card CCC0101 to use the CSUBV table. CSUBV is entered in British units only.

W1(R) Normalized flow area or normalized stem position (See Section 8.15.10 for motor valve and Section 8.15.11 for servo valve).

W2(R) Forward CSUBV $\{(gal/min)/[(lb_f/in^2)^{0.5}]\}$. The CSUBV is input in British units only and is converted automatically by the code to SI units [(m$^{7/}$kg)$^{0.5}$] using a multiplier of 7.598055E-7 as the conversion factor.

W3(R) Reverse CSUBV $\{(gal/min)/[(lb_f/in^2)^{0.5}]\}$.

The user is cautioned that when CSUBV = 0.0, the valve acts like a check valve and no flow occurs through the valve even if the normalized stem position is 1.0 or open.

8.16 Pump Component, PUMP

A pump component is indicated by PUMP for Word 2 on card CCC0000. A pump consists of one volume and two junctions, one attached to each end of the volume. For major edits, minor edits, and plot variables, the volume in the pump component is numbered as CCC010000. The pump junctions are numbered CCC010000 for the inlet junction and CCC020000 for the outlet junction.

8.16.1 Pump Volume Geometry, CCC0101-0107

This card (or cards) is required for a pump component. The seven words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R) Area (m^2, ft^2).

W2(R) Length (m, ft).

W3(R) Volume (m^3, ft^3). The program requires that the volume equals the area times the length (W3 = W1•W2). At least two of the three quantities, W1, W2, W3, must be nonzero. If